

RoboReward-8B FPS Sensitivity Analysis: Complete Results  
Task: Rearrange Pillows on Sofa (DROID Dataset)

| FPS           | Accuracy (%) | Precision (%) | Recall (%) | F1 Score (%) | Specificity (%) | TP | FN | FP | TN |
|---------------|--------------|---------------|------------|--------------|-----------------|----|----|----|----|
| default (2.0) | 23.08        | 0.0           | 0.0        | 0.0          | 60.0            | 0  | 50 | 0  | 15 |
| 2.0           | 23.08        | 0.0           | 0.0        | 0.0          | 60.0            | 0  | 50 | 0  | 15 |
| 5.0           | 26.15        | 100.0         | 4.0        | 7.69         | 100.0           | 2  | 48 | 0  | 15 |
| 10.0          | 26.15        | 100.0         | 4.0        | 7.69         | 100.0           | 2  | 48 | 0  | 15 |
| 15.0          | 29.23        | 100.0         | 8.0        | 14.81        | 100.0           | 4  | 46 | 0  | 15 |
| 20.0          | 44.62        | 100.0         | 28.0       | 43.75        | 100.0           | 14 | 36 | 0  | 15 |
| 30.0          | 56.92        | 92.31         | 48.0       | 63.16        | 86.67           | 24 | 26 | 2  | 13 |
| 60.0          | 75.38        | 86.96         | 80.0       | 83.33        | 60.0            | 40 | 10 | 6  | 9  |

Total Samples: 65 (50 Successes, 15 Failures) | Best Performance: 60 FPS (highlighted in green)